



A COMPREHENSIVE STUDY ON BLOCK CHAIN TECHNOLOGY IN SUPPLY-CHAIN MANAGEMENT

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ABSTRACT

Block chain is the basis of all cryptocurrency in the world like Bitcoins. Block chain is used for transactions of bitcoins globally without the use of any third party medium. As the internet is used for global decentralized communication, in the same way block chain is used for decentralized transactions globally. It helps in saving the supplier and consumer ends from third party involvement, tolerance network, prevention of data manipulation, and reduction in operational costs. It was first invented in 1991 and later reintroduced to the world in 2008 by Satoshi Nakamoto and now after a decade later with block chain 2.0 it is expected to be not only used in cryptocurrency but also in electronic industry in the supply chain management, electronics manufacturing industry and power and energy system. Block chain is considered as the new form of information technology that could revolutionize technology, industry, and commerce and make it a transparent, safer and reliable place. Blockchain's has a limitless potential for record-keeping in decentralized form. From greater user privacy and heightened security to lower processing fees and fewer errors, due to less human intervention, this technology can be introduced in many applications in the future.

Keywords: Blockchain, Supply Chain Management.

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1. INTRODUCTION

According to the authors of "Blockchain Revolution," Don & Alex Tapscott, "The blockchain is an incorruptible digital ledger of economic transactions that can be programmed to record

not just financial transactions but virtually everything of value”[1]. A blockchain is a chain of block in literal sense that are time-stamped and contain numerous records of data which is present with every user of that blockchain. A blockchain can be defined as a public database which contains all the digital information and these digital information is protected by mechanisms such as time-stamped of each information and each transactional data being present in every node[2].

Blockchain technology was initially developed for decentralized transactions globally in the form of bitcoins but with advancement of technology and new innovations blockchain technology is expanding rapidly from the fields of cryptocurrency to the fields of consumer electronic products, supply chain management and monitoring, identity verification and even fraud management etc.[3]. It is decentralized as all the data is not present with a single authority and the transactions are verified by a number of computers thereby increasing the transparency, accountability and incorruptible records of transactions between the suppliers which are the providers of base material and the consumers which are the end product users so that the manufacturing companies of the products are not fully accountable and not held responsible if the end product doesn't have a desirable result [4]. Some of the features of blockchain technology are as follows:

- Decentralized transactions not relying on the third party's trust
- Resilient networks due to mutual data and system management
- Previous data records cannot be tampered with because they are time stamped and only new entries can be added
- Adding of new entries or any transactional requires the consensus of every party present in the blockchain
- Infrastructure costs and operational costs can be substantially reduced

Blockchain technology may or may not be the future of internet but it sure is the future of all the supply-chain managements as it is currently being used in companies like Walmart, Maersk, British airways and FedEx to manage their supply chains from cargo data to saving time and money on restocking processes. This paper gives a brief discussion on block chain technology and its current usage in the industries.

A basic blockchain mechanism in the case of current electronic components industry supply-chain infrastructure is as follows, with the growth of semiconductor in more and more applications, there poses a serious threat of counterfeit components which may be provided by the industrial supplier leading to use of sub-standard components resulting in financial losses and damage to brand image[5]. This situation can be resolved by using blockchain technology. Blockchain technology can be used to track and verify each step along the life cycle of the products, including from the beginning of their raw materials, the steps involved in the manufacturing process, delivery dates to warehouses and consumer's home. With the introduction of blockchain technology in the supply-chain management there is a decrease in manufacturing delay, ensuring products meet the industrial regulations, preventing counterfeit versions from being sold[6]. This will increase the accountability and transparency at every level of the supply chain as every digital information will be noted and time-stamped, leading to formation of a record that can be tracked with ease.

2. LITERATUTE REVIEW

In security and privacy on blockchain paper, blockchain is defined as the innovation for the future that has immense scope in storing information, in transactions, in performing different functions, and in establishment of confidentiality, accountability and integrity of data in the

system. Blockchain performs improvement in bitcoin by providing different methods of cryptography and cybersecurity.

In the paper titled, “An Overview of blockchain technology: architecture, consensus, and future trends”, the basis of bitcoin is defined as blockchain. Blockchain is the fundamental foundation of transactions to be done in a decentralized manner. It also describes various field of applications of blockchain such as Internet of Things (IoT), financial sector, reputation sector and so on.

In [6] the applications of blockchain beyond the use in the digital transactions is defined. The blockchain is compared with the latest technology. The blockchain is explained as the foundational technology for the future not as a disruptive technology that has immense global and local reach. And its impact in various fields such as cloud, AI and IoT.

3. DISCUSSION

3.1. Blockchain Basic Concepts

Hence, Blockchain meticulously documents every transaction and distributes that cryptographic data across decentralized nodes[7]. It is a decentralized database that uses validation for decision making by consensus algorithm and security preserved by encryption methods. It is a digital library which keeps records of all information in the form of blocks which are encrypted and the records are linked to one another in the form of chains after validation through certain certifications to avoid tampering with the data of digital documents. Each block is connected to the previous block to form a chain of records called a Blockchain [8].

3.1.1 Types of Blockchain

General Blockchain

This is considered as an open-source and decentralized blockchain in which anyone can have access to the information of the transaction and can participate in the transaction, consensus process and voting. This is also referred to as an open chain and is used for enterprise purposes.

Private Blockchain

This usually belongs to a specific person such as personal records, medical records, or confidential company records. This kind of blockchain is not decentralized as the transactions are reversible, the validators are authenticated there making the consensus algorithm run faster. This kind of blockchain is suitable for database management, inspection and auditing and other internal processes of an organization.

Consortium Blockchain

The permission for validation of the blockchain is given to certain nodes only making it an almost decentralized network.

3.1.2. Proof

Blockchain is a system in which there is not centralized trusted authority and it requires a consensus method for its nodes as a validation method and doesn't require a third party to verify it. For any transaction to be validated it should be approved by 51% of nodes in the blockchain network. There is different consensus method followed in a blockchain like:

- Proof of work
- Proof of stack
- Proof of validity

The mechanism followed by blockchain to maintain security of the block, to keep it decentralized are as follows:

- Hashing mechanism
- Proof-of-work
- Peer-to-peer network such that everyone gets a full copy of the blockchain.

Here we will discuss the blockchain mechanism used in the supply-chain management or electronics management services. Blockchain is used in the manufacturing industries or in the supply chain management due to high friction between the parties involved, when the manufacturing of a single product is a multi-party process, and when there is lack of trust between the companies involved[9].

3.1.3. Blockchain helps in resolving the above issues by the following process

Supplier contract management

Smart contracts can be introduced to improve transaction efficiency, bring transparency in the process thereby increasing the execution speed of the process, helping in faster conflict resolution and eliminating the hassle of maintaining a paper contract.

Digital thread

It is used for the manufacturing of products where multiple companies are involved as it increases the trust between the companies by developing a communication framework that connects elements in the manufacturing process and provides an integrated asset view.

Production tracking

This helps the manufacturing companies in keeping track of products completed by their electronic manufacturing services in real time resulting in improved authentication and real-time tracking of products.

Tracking asset maintenance

Blockchain helps in tracking of the asset where multiple parties are involved.

Tracking recalled products

Blockchain keeps track of any individual product's journey from its manufacturing, assembling and its completion till it reaches the consumer.

3.2. Blockchain in manufacturing supply chain management

The following are the steps [10]involved in the supply chain management in blockchain

- Interface involved in the blockchain system
 - A mobile app for company and merchants
 - Website for suppliers, manufactures and QA team
- Nodes that are present in the blockchain
 - Merchants - who placed the order
 - Manufacturing Company - manufactures the product
 - Suppliers - raw material provider to the manufactures
 - Logistics Company - involved in the distribution of the product
 - Quality Assurance Team - for quality check of the product
- Merchants place the order by signing in the website and filling the forms of product requirements. When the information is filled by the merchant a block is formed which contains timestamp, hash value, previous hash value, and other digital and transactional information.

- The manufacturing company receives the order after which the manufacturers can access the stored data in the block by the hash address from the merchant and after getting the product requirement the manufacturing company makes a Bill of Material (B.O.M).
- Different suppliers are contacted to get the raw materials for the product. And as soon as the details and type of material are updated in the blockchain smart contracts are made and suppliers are notified about the new order.
- The supplier receives the new order of the raw materials from the manufacturing company through the blockchain. Now, the suppliers add the invoice with the images of the raw materials in the blockchain to update all the stakeholders or nodes of the blockchain.
- Thus, new information added by the supplier is validated by the nodes of the blockchain the next step is to deliver the raw materials to the manufacturer and adding new updates to the information in the blockchain like the type and quantity of the order.
- After the order is packed by the supplier a logistics service provider company is contacted to deliver the order to the manufacturer and the status is updated in the blockchain.
- Logistics service provider adds the details of the delivery, dispatch time, driver name, vehicle number, number of cartons received from the supplier and type of products on the blockchain.
- Nodes of the blockchain network can ensure that the number of received products is the same as that of the dispatched product by the supplier by accessing the stored information on the blockchain.
- When the order is received by the manufacturer then they quality of the material is checked and if the quality of the components is not up to the mark of the requirement then a recall request is sent to the supplier otherwise, when the quality of the material received is good then then the payment is made directly to the supplier in their account according to the regulations of the smart contract.
- All the transactions details are recorded in the blockchain and its copy is sent to all the nodes of the blockchain network. Once the goods are received by the manufacturer and the quality of the delivery is checked then the products are handed over to the QA team to do a quality check.
- After the quality check the QA team adds the details on the blockchain manufacturing platform. The following information is added to the blockchain:
 - Methods implemented while testing the product
 - Product guidelines
 - Tests' results
 - Product certification
- The next step is to deliver the final product to the merchant. The manufacturer adds the invoice of the product's manufacturing cost and its shipment details to the blockchain which will be visible to the merchant. This will be visible to other stakeholders and the blockchain network will help in bringing fairness and trust in the system.

3.3. Advantages of blockchain

3.3.1. Accuracy of the chain

Transactions on the block chain are verified by the network of thousands and millions of computers which are connected to the blockchain. Thereby, decreasing the human intervention in the process and increasing the accuracy by decreasing the human error. Any transaction is spread across the rest of the blockchain only if it is verified by at least 51% of the network's computers.

3.3.2. Operational cost reductions

Blockchain reduces the involvement of a third party in any process like transactions of money globally in the form of bitcoins which previously had to be done through a bank and took a lot of time and extra money.

3.3.3. Decentralization

Blockchain does not have a central authority to store its digital information. Instead the information is copied on the entire blockchain and spread across a network of computers. By spreading the information across the network rather than storing it in one central database, blockchain becomes more difficult to tamper with.

3.3.4. Transactions

Transactions are efficiently completed in about 10 minutes and considered secure after just a few hours. All transactions made are confidential and the identifying details about the users making the transactions are kept private. The authenticity of every transaction is verified by the entire blockchain network. Hence, transactions are secure, private, efficient and traceable.

3.3.5. Transparency

Blockchain being an open source technology means that users can modify the code as long as they have a majority of networks computational power backing them. The start of the blockchain is always a single source.

3.4. Disadvantages of Blockchain

3.4.1. Technology Cost

The cost of the amount of electricity used for the validation of transactions in blockchain is close to what Denmark consumes annually. The technology cost includes the infrastructure cost, electricity cost and mining costs which leads to more and more energy consumption. Hence, Infrastructure becomes expensive with additional technological costs.

3.4.2. Speed Inefficiency

In blockchain it takes about 10 minutes to add a block to the existing network of the blockchain.

3.4.3. Hack Susceptibility

The newer crypto currencies and blockchain network are susceptible to 51% attacks as attackers now simpler rent the computational power rather than buying all the equipment.

4. APPLICATIONS OF BLOCKCHAIN TECHNOLOGY

Blockchain technology can also be used to store digital data, other types of transactions in electronics or supply-chain industries, voting for candidates in election and information about property exchange as well as other business operations[11].

4.1. Banking sector

By integrating blockchain in the banking sector there will be more transparency and faster operations in the system even if the volume of transactions that the bank needs to settle are plenty. It will take approximately 10 minutes for each transaction to complete despite the date or time of deposit.

4.2. Healthcare industry

The patient's medical record can be put on the blockchain and can be accessed by other doctors to see the patient's medical history with a private key so that disease detection becomes easy.

4.3. Property records use

Registering property deeds in the blockchain so it can be verified by the government employees, local recorder's office which will make the process efficient, less time consuming, and less expensive.

4.4. Used in smart contracts

Smart contracts is a self-executing contract as the terms and conditions of the contracts are automatically enforced by the ledger itself. Smart contracts make the business model more efficient, profitable, efficiency and security.

4.5. Use in supply chain

Blockchain helps to streamline the end-to-end supply chain process and possibly save both time and money in the process. As blockchain keeps records of the entire process of components manufacturing, assembling and final marketing of the product making the records digitally permanent, audit-able which making everyone accountable at every step.

4.6. Use in voting

Using blockchain in elections will eliminate the election frauds and boost voters turn out as each vote will be stored as a block in the block chain making it impossible to tamper with.

5. CONCLUSION AND IMPLICATIONS

The use of blockchains that was started from cryptocurrency has found its use in the manufacturing, retail and customer based markets too. With blockchain technology the trust and the fairness in the system is maintained as not only it makes the system transparent to all the stakeholders of the process but also leads to the accountability of the person responsible for the task. It leads to a better governance of the system without any third party intervention. It decreases the operational cost of the system and due to the mechanisms present in the blockchain technology it leads to decrease in data tampering. Blockchain can help in providing better supply chain visibility, managing complexities in the supply chain of original electronic manufacturing and manage complexities of electronics value chain. With the level of innovations in technology it's only a matter of time when all the companies will switch to blockchain technology for all its process implementation and blockchain will become a key technology in bringing mammoth transformations in the industry. Blockchain allows transactions to take place in decentralized manner. With future innovations in the technology it has immense scope in the field of storing information, transaction executions and establishing integrity and confidentiality in an open environment.

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